$$r_1 = F_{:,1}/\|F_{:,1}\|, \quad r_2 = \frac{F_{:,2} - (r_1^\top F_{:,2}) r_1}{\|F_{:,2} - (r_1^\top F_{:,2}) r_1\|}, \quad R = [r_1, r_2, r_1 \times r_2] \in SO(3)$$